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Space as a Parameter in Electroacoustic Music: A critical analysis of Denis Smalley's Spatial Articulation

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Abstract

The proliferation of spatialisation technology has encouraged a mass of unclassified techniques for the use of spatial manipulation in electroacoustic music. The complexity involved with theorizing the composers conception, the transmission of sound via electroacoustic means and the resulting listeners perception of a spatial work is a multifaceted and interdisciplinary task.

Composer and theorist Denis Smalley in his essay *Spectro-morphology and Structuring Processes* proposed arguably the most influential theory for sound spatialisation in electroacoustic music of the last three decades. The focus of this study is a critical analysis of Smalley's theoretical framework for *Spatial Articulation*, and the contrast of these findings with recent developments in advanced spatialisation technology and successive theories.

The study consists of a literature review of proposed theoretical frameworks relating to and including Smalley's Spatial Articulation.

Declaration

The candidate declares that the material contained in this submission is his/her own work and that appropriate recognition has been given when referring to the work of others.

Signature:.....

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Date:.....

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Introduction

Since the advent of the loudspeaker musicians have developed greater degrees of control over sound. This proliferation of electroacoustic technologies available to the musician has created an abundance of approaches to deal with sound material. As a result of both cheap and powerful computation and the convergence of the technical disciplines with music, the degree of control over spatialised sound has become more acute.

Advanced processes and techniques for sound spatialisation such as wave field synthesis and ambisonic projection have enabled the composer of electroacoustic music to more accurately represent space in an expressive way. However there lacks a cohesive theoretical framework for electroacoustic spatialisation that is inclusive of both these new technologies and the many aspects of auditory perception.

Composer and theorist Denis Smalley in his essay *Spectro-morphology and Structuring Processes* (SSP) proposed the notion of Spatial Articulation, arguably one of the most influential theories for sound spatialisation in electroacoustic music of the last three decades. In this essay Smalley has specifically addressed spatial perception in electroacoustic music composition primarily from a phenomenological perspective. Spatial Articulation is a commonly cited tool used for the exploration of space as an expressive entity in electroacoustic music. The focus of this study is a critical analysis of Spatial Articulation, and the contrast of these findings with successive theories influenced by recent developments in advanced spatialisation technology. In order to offer a critique of Spatial Articulation, an outline must first be drawn of the theoretical style of SSP. The choice of Spatial Articulation, and by extension, SSP as the base sample of this study is justified in its direct lineage to established electroacoustic principles posited in the tradition of Schaefferian theory and the clarifications of Michel Chion.

Although there exists a rich history of spatial considerations in instrumental music[1] the focus of this paper will reside within the sphere of electroacoustic music. This is necessary to delimit the subject material to an appropriate size for this study.

In this paper the term 'electroacoustic' will be used to denote the act of producing or reproducing sound by the conversion of electrical energy to acoustic energy. This term is commonly used to define music whose only source of sound is the loudspeaker.

This study will be adopting Smalley's perspective of addressing electroacoustic music that is acousmatic in nature, that is music whose sounding origins are not visible. As such the term 'electroacoustic music' in this context refers to music that exists within the recorded format. This distinction must be made to bracket considerations of live performance and musicianship of which the scope of this paper will not be able to address.

The term 'Space' and by extension 'Spatialisation' is important to define as these words have a diverse range of meanings within the fields of composition[2], musicology and sound engineering. In reference to the subject matter the use of the word spatialisation will broadly to encompass any articulation of space as an aspect of electroacoustic music. This distinction will be further clarified during the analysis of proposed theoretical frameworks as each redefines the specific meaning of spatialisation to fit their purpose.

The acronym SSP will be used to refer to *Spectro-morphology and structuring processes* as a whole as opposed to spectro-morphology proper, which is defined as an aspect of Smalley's overall theory relating to the spectrum of the sound (spectra – frequencies and amplitudes) and the way in which it changes over time (morphology)[3].

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[1] Maria Anna Harley (Maja Trochimczyk). 1994. *Space and Spatialization in Contemporary Music: History and Analysis, Ideas and Implementations*. Ph. D. Dissertation. Montreal, Quebec, Canada: McGill University, Faculty of Music. PDF Reprint. Los Angeles, California: Moonrise Press, 2011.

[2] Worrall, David. "Space in sound: sound of space," *Organised Sound* (1998), Accessed September 2, 2014. http://journals.cambridge.org/abstract_S1355771898002027

[3] Christian Haines, "Sound Organisation (1)," (Seminar presented in course, 'Sound Design', University of Adelaide, Elder Conservatorium, August 3, 2012).

Chapter 1 - Smalley's adoption of Schaeffer's Phenomenology

Spectro-morphology and structuring processes (SSP) first appeared in a paper for the EMS[4] electroacoustic music conference in Stockholm 1981 and was later published in a collection of essays entitled *The Language of Electroacoustic Music*[5]. Here the chapter resides amongst other influential writing from Pierre Boulez, Trevor Wishart, Tod Machover, Barry Truax and Simon Emmerson. Editor Emmerson describes the purpose of this collection by highlighting the similarities in approach taken by the aforementioned contributors. Emmerson first refers to their "…commitment to experimental tradition, one which seeks to find new solutions, to test, to research and to modify [musical] assumptions accordingly"[6] and secondly, a reference to these composer/theorists recognizing a "…commitment to communication and an acceptance of the proper role of the listeners perception in the musical process."[7] Smalley's chapter is placed under the sub section of Materials and Language in which he defines spectro-morphology as an approach to addressing electroacoustic music "...based entirely on the way sound-objects have, inherent within their perceptual qualities, the potential for building gestures, shapes and forms[8]"

In general SSP provides a descriptive tool set based on aural perception that is designed to encompass the drastic expansion of sonic possibilities afforded by the development of electroacoustic technology. Although not strictly a compositional theory or method, Smalley posits that SSP can be used to influence compositional methods by helping the composer learn a system of concepts and words to help diagnose and describe sound in all its facets[9].

Recognising Smalley's focus on the importance of the listener's perception provides insight into the development of SSP. Smalley emphatically states that a focus upon aural perception

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[4] Elektronmusikstudion
[5] Denis Smalley, "Spectro-morphology and Structuring Processes," in *The Language of Electroacoustic Music*, ed. Simon Emmerson (Basingstoke: Macmillan Press, 1986)
[6] Simon Emmerson, "Introduction," in *The Language of Electroacoustic Music,* ed. Simon Emmerson (Basingstoke: Macmillan Press, 1986).
[7] Emmerson, 2
[8] Emmerson. 3.
[9] Denis Smalley, "Spectromorphology: explaining sound-shapes," in *Organised Sound* (1997), accessed April 27, 2015, http://journals.cambridge.org/abstract_S1355771897009059

is of primary importance in the apprehension of electroacoustic music, and that musical apprehension is based on the finding of a "perceptual affinity with [music's] materials and structure."[10] It is this inherent idea that justifies the construct of SSP, in particular as an alternative to the abstract formalism of early twentieth century western music theory or the reliance upon non-musical models to support a composer's conception.

A defining feature of Smalley's theory is the influence of Pierre Schaeffer's phenomenological reduction of sound through the practice of reduced listening. Schaeffer's contributions are oft cited in text with a majority of other reference material either directly or indirectly grounded in Schaefferian theory.

In a critical analysis of Schaefferian theory, Brian Kane highlights the phenomenological grounding of Schaeffer's new classification of sounds as the *sound object*. Here Kane shows Schaeffer's theoretical reasoning by using a simplified distinction of Husserlian phenomenology. Specifically Kane shows in his critique that a division of thinking between what he terms *realism*[11] and *psychologism*[12] is essential to achieve the phenomenological reduction that Schaeffer undertook. Realism in Kane's words is concerned with the consideration of factual objects that have an accepted basis in the physically measureable world. Further Kane uses the Husserlian terminology of 'the natural standpoint' to ground realism.

> In the natural standpoint we faithfully assume ourselves to be subject situated within the external world; we take reports of our senses to be causal effects of externally existent

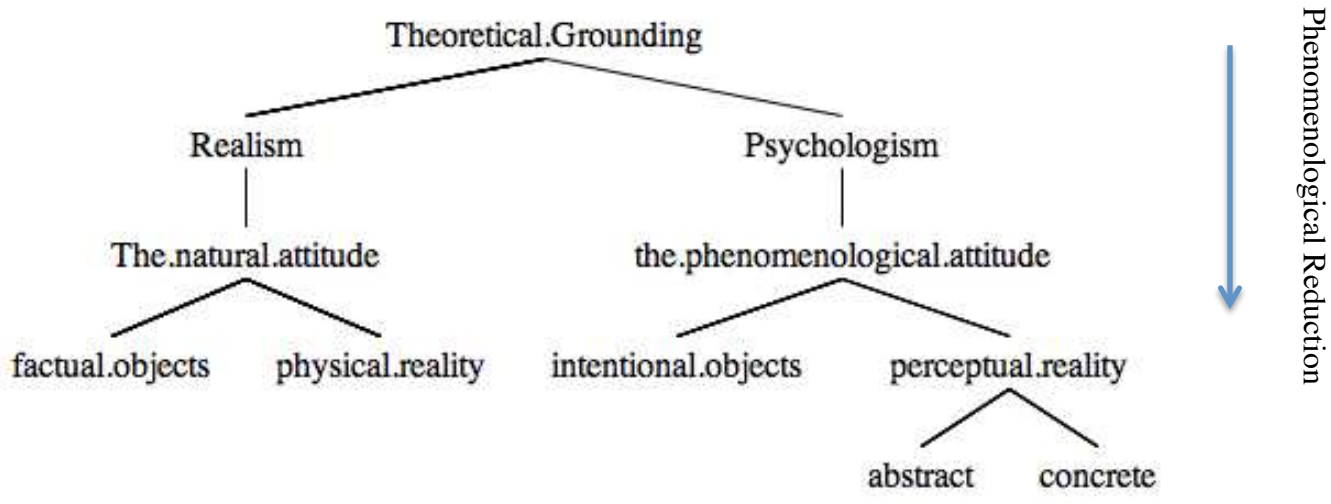
[10] Smalley (1986), 62
[11] Generic Realism: "*a, b* and *c* and so on exist, and the fact that they exist and have properties such as *F-ness, G-ness* and *H-ness* is independent of anyone's beliefs, linguistic practices and conceptual schemes, and so on" - definition at Stanford encyclopaedia of philosophy.
[12] Brian Kane, "L'Objet Sonore Maintenant: Pierre Schaeffer, sound objects and the phenomenological reduction," in *Organised Sound* (2007), accessed May 17, 2015, http://journals.cambridge.org/abstract_S135577180700163X.

things. Any claims that posit physical subsistence to exterior objects depend on the natural standpoint, and this includes the claims of the physical sciences.[13]

In contrast Kane defines psychologism as concerned with the intentional object or the perceptual content induced in the human psyche. In this context another word for this perceptual content is the phenomenon.

In contrast to Kane's assessment of Schaeffer's reasoning, Kim addresses this definition as an erroneous use of the term psychologism, a term which Husserl supposedly reassesses in later writing. Here alternate terminology is used to describe this dualism in thought, or two differing attitudes towards experience and knowledge, the *natural attitude* and the *phenomenological attitude*[14]. The former representative of the natural standpoint and the latter the phenomena induced in psyche.

As below the process of phenomenological reduction is demonstrated with regard to sound.



A prerequisite for engaging with a theory of the perceived phenomenon is this phenomenological reduction, or in Kane's words, "the general abstention from theses bound to the natural standpoint."

[13] Kane, 17
[14] Suk-Jun Kim, "A Critique on Pierre Schaeffer's Phenomenological Approaches: Based on the Acousmatic and Reduced Listening," (paper presented at the Pierre Schaeffer Conference: mediART in Rijeka, Croatia, on Oct. 7, 2010), http://reddoorsound.com/files/PS_Phenomenology_KIM.pdf

Schaeffer's reductionist approach can be seen to have greatly influenced both Smalley's use of language and terminology and therefore his theoretical reasoning. As such Smalley's approach can also be considered reductionist[15]. A primary concept employed in Smalley's theorisation is that of the continuum between perceiving the *abstract* and the *concrete* nature of sound, an idea arrived at through Schaefferian reduction and which remains a foundation of his thinking.

> All sounds possess this dual potential – the *abstract* and the *concrete* aspects of sound – and all musical structures are balanced somewhere between the two, although exactly how they are found to be balanced can vary greatly among listeners. This is because all listeners have considerable practice at the concrete aspect in daily life, while an abstract approach needs to be acquired…Balancing abstract and concrete attitudes is therefore a question of both competence and intention.[16]

Michel Chion provides a description of the *abstract/concrete* nature of sound and its dual potential in an influential interpretation of Schaeffer's *Traite des objet Musiceaux*. Chion shows Schaeffer's use of a definition for *abstract* and *concrete* proposed by French philosopher Andre Lalande. Here Chion describes the nature of *abstract* and *concrete* as "the two isotopes of reality, the two faces of every perception, interdependent and complimentary, which must be reconciled in music, against the excess of *concrete* (in 'savage' musique concrete) or the excess of abstract (in serial and other types of a priori musics)."[17] As such Smalley's terminology reflects this interpretation[18].

Chion famously re-articulated and clarifies Schaeffer's proposition of the existence of different listening modes. The importance placed in reduced listening to achieve an abstraction of sound material is clear and in essence this is the method by which to undertake

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[15] Natasha Barrett, "Spatio-musical composition strategies," in *Organised Sound* (2002), accessed April 3, 2015, http://journals.cambridge.org.proxy.library.adelaide.edu.au/article_S1355771802003114, 313.

[16] Smalley (1986), 64

[17] Michel Chion, *Guide des objets Sonores: Pierre Schaeffer et la recherché musicale*, trans. John Dack and Christine North (Paris: Édition Bucket/Chastel, 1983), 37

[18] Smalley (1997), 111

a phenomenological reduction of sound. As a result, the practices of listening and hearing, both the active and passive aspects of auditory perception, have been categorically defined into various modes of listening. Michel Chion provides a formalisation of this idea into the 'three listening modes'[19]. Others, including Smalley, further expand this idea to systematically define specific aspects of auditory perception and their resulting phenomena[20]-[21]-[22]. This type of categorisation is an example of the lasting influence of Schaefferian theory upon electroacoustic music.

As we have seen, the use of reduced listening to *abstract* and investigate sound in and of itself forms a foundation for SSP, however equal acknowledgement is given to the musical power of the *concrete* sonic reference and mimesis of sound. Smalley also embraces this referential aspect of perceptive musical meaning within his theoretical framework.

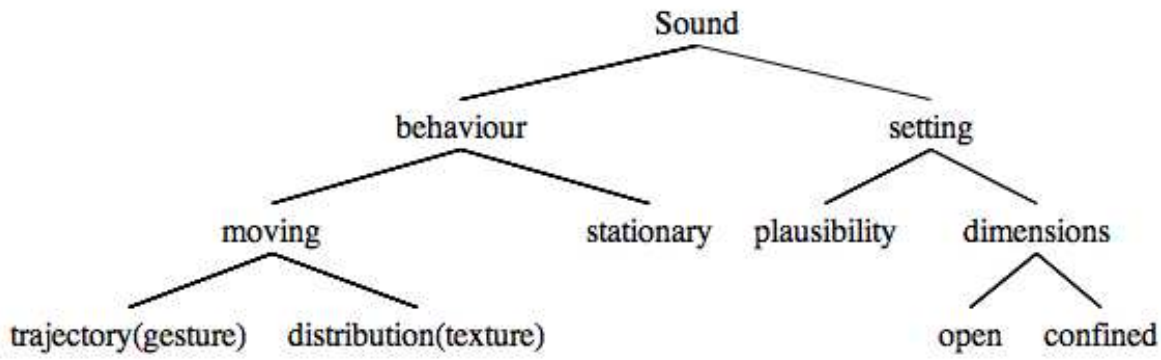
When spatialisation of sound is considered in SSP, the acceptance of this dualism becomes central to Smalley's theorization. As such, a critical analysis of the properties of *Spatial Articulation* will highlight how this methodology is used to address the nature of space as an expressive aspect in acousmatic electroacoustic music.

[19] Michel Chion, *Audio-Vision: Sound on Screen*, Columbia University Press, 1994, 25-34
[20] Smalley (1997), 108-109
[21] Warren Burt, "Ways of Listening," accessed May 28, 2015, http://www.warrenburt.com/ways_of_listening/
[22] K Tuuri, T Eerola, "Formulating a Revised Taxonomy for Modes of Listening," in *Journal of New Music Research* (2012), accessed May 28, 2015, http://www.tandfonline.com.proxy.library.adelaide.edu.au/doi/citedby/10.1080/09298215.2011.614951#.VYdtncYsojI.

Chapter 2 - Denis Smalley's Spatial Articulation

Smalley defines Spatial Articulation as concerning the "interaction of sound structures with the properties of the acoustic environment it inhabits"[23] in an aim to help composers utilise spatialisation to articulate musical structure.

The properties of spatial articulation in an electroacoustic work can be divided into two distinct paths of consideration. The first is that of a sound (or sound structure)'s *behavior*, an idea intrinsically linked with sound localisation in psycho-acoustic terminology. This *behavior* of sound is determined by whether sound is *stationary* or *moving*, with movement further articulated in two ways: as a *trajectory* (a spatial representation of gesture) or as *spatial distribution* (a spatial representation of texture). The second distinct path of consideration is that of the *setting* of sound structure, which is then separated to the *dimensions* and the *plausibility* of spatial setting. The dimensions of a setting are characterized by whether this setting is indicative of an *open* or a *confined* environment, while plausibility is judged by a spatial settings representation of either a *realistic* or *fictitious* space[24].



:: Smalley's properties of Spatial Articulation::

[23] Smalley (1986), 90
[24] Smalley (1986), 91

Smalley's adoption of Schaeffer's phenomenological reduction is established when the notion of *abstract* and *concrete* sound perception is echoed in the outline of the properties of spatial articulation. In particular the initial division into two paths of consideration, behavior and setting, can be considered akin to Schaeffer's original reduction to a dualism of *abstract/concrete* sound by the application of reduced listening. As such the *trajectory* or *distribution* of a sound assumes an *abstract* conception a sound as a sound object, whereas the *dimensions* or *plausibility* of a setting must be conceptualized in a referential or *concrete* way.

Smalley posits that the *transference of composed spatial articulation* is an important aspect associated with *spatial articulation* proper. This *transference* refers to the actual projection of music into the listening environment as opposed to the controlled composition environment of the studio. The neutrality of both the technology used to project sound and the eventual listening space is called into question and as such both are considered to affect the musical substance and structure of a work. This acknowledgment of *transference* is described to help the adaptation of *spatial articulation* in a work to the idiosyncrasies of a particular listening environment. In most electroacoustic music discourse this ad hoc process is named *sound diffusion*[25].

Of interest to note is how Smalley has deemed this as an integral part of the composer's responsibility within his theoretical framework. Most traditional understandings of this *transference* are that it encompasses a role to be undertaken by specialists outside of the field of composition. Analogies can be made to the sound engineer and the orchestral conductor in this respect.

In his initial proposition, *spatial articulation* is concerned primarily with the presentation of spatial concepts *through* loudspeakers and *into* listening spaces as opposed to the more stable environment of headphone listening. This important distinction is made explicit in later writings.

[25] Smalley (1986), 92

2.1 – The further development of Spatial Articulation

A decade after the publication of SSP Smalley undertakes a comprehensive rewriting of his original theory, expanding upon core principles in his perceptual framework and refining its use of language. Of pertinence to the consideration of space in electroacoustic music is the further development of the *abstract/concrete* perceptual dualism proposed by Schaeffer. In these writings this idea is reframed with new terminology borrowed from musical semiotics.[26]

The term *Intrinsic* becomes the successor to Schaeffer's reductionist *abstract* 'sound object' in and of itself, and here explicitly states that spectro-morphology is concerned with *intrinsic* features, that is "…[spectro-morphology] is an aid to describing sound events and their relationships as they exist within a piece of music."[27] Therefore *Extrinsic* is hereafter the successor to the *concrete* mimetic and referential aspect of auditory perception. A further clarification is made of the acceptance that music can rely on the relation to experiences outside of the context of a composition, and as such can be considered a cultural construct, therefore an "…*extrinsic* foundation in culture is necessary so that the *intrinsic* can have meaning."[28] The invention of the term *source bonding* describes the link between intrinsic-extrinsic aspects in electroacoustic music. Smalley defines *source bonding* as "the natural tendency to relate sounds to supposed sources and causes, and to relate sounds to each other because they appear to have shared or associated origins"[29]

In this context another development of the definition of reduced listening is proposed to clarify the adoption of the modified language. Here reduced listening is contextualized as an

[26] Jean-Jacques Nattiez, *Music and Discourse: Towards a Semiology of Music* (Princeton: Princeton University Press, 1990), 102-129.
[27] Smalley (1997), 110
[28] Smalley (1997), 110
[29] Smalley (1997), 110

"…abstract, relatively objective process, a microscopic, intrinsic listening"[30] essential to spectro-morphology.

The updated purpose of Smalley's consideration of space is to define a grammar of localisation with which one can facilitate changes in spatial perspective as a means of delineating musical structure. Smalley's uses the term localisation in a general sense to encapsulate both aspects of spatial auditory perception as defined by Schaeffer's phenomenological reduction (*abstract/concrete* or *intrinsic/extrinsic*), and its new synthesis as his spatial articulation. This is an idiosyncratic use of the term, as opposed to its traditional usage in psychoacoustics relating specifically to quantifiable, parameter-based localisation of sound.[31]

The actual properties of spatial articulation are comprehensively expanded to encompass a range of descriptive sub-definitions, however the underlying structure of the original theory remains essentially unchanged. These sub definitions are descriptively diametric in nature, examples of such language include, open/free, shallow/narrow, opaque/blurred-transparent/clear, definition/fill. As such Smalley inadvertently obscures his initial framework by offering ever more subjective descriptors of space. The final set of guidelines offered in this revision encompasses the definitions of an overall *spatial style*, seemingly the only new offering to his original conceptions for space in electroacoustic music. These global *spatial styles* are described by idiosyncratic syntheses of the properties of spatial articulation that describe the overall spatial setting of a work as a whole. This analytical descriptor allows the development of what Smalley calls *spatio-morphology*. From the perceptual point of view diagnosing a change of spatial perspective by defining changes in the overall spatial style

[30] Smalley (1997), 111
[31] Andy Farnell, *Designing Sound*, (The MIT press, Cambridge-Massachusetts, 2010), 79.

allows us to perceive a transformation of space, as opposed to change in the *intrinsic* spectro-morphology, as such "…the aural interpretation of space attains a new structural status."[32]

Smalley again highlights how the instabilities of grammatical details and psychological messages relating to spatial apprehension are exposed when super positioning the composed space within the listening space.

An interesting point is the brief recognition to that fact that "…real spaces can now be simulated faithfully."[33] This shows an acknowledgment of advanced spatialisation technologies[34], a field in which the accurate reproduction of naturally occurring physical wave fields form the basis in the development of electroacoustic technology. This brief reasoning from the realist, 'natural standpoint', illuminates one of few theoretical departures from a firm phenomenological grounding in spatio-musical auditory perception.

During the period between his original theory and his comprehensive revision, the focus of Smalley's auxiliary writings are to support a focus on the perceptual aspects of music experience. His contributions include succinct explanations of spatial experience in electroacoustic music from the perspective of the listener and as such formulate prescriptions for the composer's responsibility to this[35]. These explanations summarize concepts developed earlier, and in doing so take a more general standpoint by considering how these theoretical frameworks for spatial experience can be applied to electroacoustic music composition. Here a far greater emphasis is placed on the problematic nature of adapting a carefully composed musical space from the studio into an unpredictable listening space. In one these summarizations from his later writing, Smalley again makes reference to the fragility of the

[32] Smalley (1997), 124
[33] Smalley (1997), 122
[34] Nathan Abbott, "Facilitating Spatial Music: assessing feasibility in the amateur implementation of advanced spatialisation technology," (Undergraduate essay, University of Adelaide, 2014), 3.
[35] Denis Smalley, "Spatial Experience in Electroacoustic Music," in *L'Espace du Son II*, ed. Francis Dhomont (Ohain: Musiques et Recherché, 1991), 123, accessed April 8, 2015, http://www.musiques-recherches.be/images/stories/eds2.pdf

transference of spatial idea's into the listening space and explicitly states that "We have yet to arrive at a state where the space-field has become totally holosonic, or at least where such technological control of the space field becomes a practicable proposition for composers."[36]

Again in this brief entrance into the diametrically opposed theoretical grounding of realism, or in the natural attitude, Smalley highlights an acceptance of the physically measurable nature of the space-field. Smalley has hitherto abstained from engaging in theory relating directly to electroacoustic technological development. By adopting the phenomenological basis in electroacoustic music theory first championed by Schaeffer and later refined by Chion, Smalley's theoretical reasoning can be seen to have passively rejected influence from the sphere of the quantifiable parameter in order to focus on the importance of aural perception and the perceived phenomena, that which mediates the musical affinity between composer and listener.

From one perspective it can be said that SSP has succeeded in developing a theoretical framework that systematically approaches perceptual analysis and compositional considerations in electroacoustic music[37]. One of Smalley's many achievements in this theorisation is the cohesive realignment of Schaeffer's reduced listening with other communicative and indicative modes of listening to show the importance of both the *intrinsic* and *extrinsic* value of sound in an electroacoustic context.

Critics have regarded this style of systematic classification in SSP as "prohibitive to practical application.[38]" However its great benefit as a guide to the perception of electroacoustic music experience is clear. It is when spatialisation is considered in this theoretical discourse,

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[36] Denis Smalley, "The Listening Imagination: listening in the electroacoustic era," *Contemporary Music Review* (2009), accessed April 8, 2015, http://dx.doi.org/10.1080/07494469600640071

[37] Tim Reed, "I. Planes of discourse in fixed media electroacoustic music: A comparative study and application of analytical approaches, and II. Three movements for string orchestra," (Ph.D. thesis, University of Florida, 2008), 34, accessed May 10, 2015, http://proxy.library.adelaide.edu.au/login?url=http://search.proquest.com.proxy.library.adelaide.edu.au/docview/250785332?accountid=8203

[38] Reed, 36

particularly in later writing in support of Spatial Articulation, that ambiguities in definitions become more prominent and ever increasingly subjective descriptors are used to support the outline of phenomena relating to spatial experience. It is possible that this has to do with the substantial acoustic variables introduced, and the fragility inherent in the electroacoustic *transference* of the *composed space* into that of the *listening environment*.

Smalley's acceptance of the primacy of aural perception has given spatialisation a sturdy platform within the last three decades of electroacoustic music discourse. The following quote encapsulates both Smalley's conclusive perspective on the use of space as a rudiment in electroacoustic music and the inadvertent rejection of the natural attitude through the adoption of Schaeffer's phenomenological reduction in forming his theory for spatial articulation.

> …since spatial experience depends on all aspects of content and context, it is the product of a collaboration of factors which cannot be conveniently packaged or easily discussed as an independent parameter. Musical spatial experience can therefore *not* be *systematically* controlled by the composer. Ultimately the composer cannot work abstractly, theoretically and technically with space, but must approach spatial awareness experientially.[39]

[39] Smalley (1991), 126

Chapter 3 - An Alternative Grounding for Spatial Music Theory

In a later attempt to reinvigorate discourse describing how spatialised sound can be used in electroacoustic music composition, Doornbusch and McIlwain[40] highlight a growing trend in theory by stating that the composition of sounds in space is considered increasingly important, but that the many previous studies in the area have been ambiguous in their presentation. This statement is further contextualised by the recognition of ubiquitous types of spatial manipulation, such as the application of reverberation, which is commonly applied as an effect and therefore is essentially seen as an afterthought. In contrast their viewpoint mirrors Denis Smalley's in that space constitutes a rudiment of electroacoustic music composition. However the reappearance of the word *parameter* within the title of their study hints at a more developed understanding of the possibilities inherent in technological developments in spatialisation.

To address this notion of the function of space as a *parameter* of composition, Doornbusch and McIlwain make a broad assumption in the delimiting of what they define as technological limitations. The ability to completely control the size and position of any source (or *sound object*) within a listening environment is taken as a given in this situation, as is the complete *parametric* control of the listening environment itself, or what is termed "the ability to control a virtual environment in all its aspects."[41] This is an effective way to focus upon semantic issues relating to space and electroacoustic music however in doing so, this approach adopts the phenomenological attitude in it's bracketing of technological consideration. This standpoint is taken for the sake of pragmatism and leaves an important link untended between the theories of how space is experienced in a musical sense, and the means by which this is achieved in praxis.

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[40] P Doornbusch & P McIlwain, "Integrating Spatial Parameters in Composition," Accessed September 5, 2014, 2 https://www.academia.edu/168451/Integrating_Spatial_Parameters_in_Composition_Practice.

[41] Doornbusch & McIlwain, 2

Others have formulated theoretical frameworks for dealing with space as a rudiment of electroacoustic music by hybridising disciplines of both the phenomenological and natural attitudes. Barrett shows an intersection between the reductionist, perceptual, phenomena centric approach of Smalley et al. with the consideration of the technological affordances made available through development in direct sound field synthesis approaches to spatialisation[42].

Barrett encapsulates many aspects of Smalley's phenomenological attitude into two distinct prescriptions for a *Spatio-musical composition strategy*, that of the auditory *illusion* and *allusion*. The application of spatial *illusion,* enacted through phantom imaging techniques[43] relates to the *sound object* and as such falls under the *abstract* categorisation of sound as defined by Schaeffer et al. This means that spatial *illusion* is directly related to the sound objects *intrinsic* properties, or in Smalley's terms, their spectro-morphology. Spatial *allusion* in Barrett's words relates to space as implied by aspects other than *intrinsic* spatial illusion. Herein lies the use of the *extrinsic/concrete* referential aspect of spatial perception where a 'listening imagination' is required of the listener who has more of an active role. In this instance the composer must make assumptions about how an *alluded to* space will be interpreted by the listener. The allusion of space is related to the nature of the source, or its the *source-bond*[44] using Smalley' terminology.

Here Barrett makes a major tangential adjustment in theoretical grounding. A third prescription is defined which relates to the direct synthesis of the three dimensional sound field. This type of spatialisation can be described as a progression away from point source

[42] Barrett, 314
[43] Phantom imaging refers to spatial manipulation in the stereophonic projection of sound using rudimentary reverberation, delay and amplitude panning. It is highly reliant upon a listeners positioning within the *sweet spo*t, a small area at one point equidistant in correlation to two or more loudspeakers.
[44] Smalley (1997), 110

speaker configurations. This includes formats such as stereophony, which must rely on illusion/allusion through *intrinsic* phantom imaging techniques and *extrinsic* associations to convey spatiality of sound.

In addressing the theoretical developments of disciplines grounded in the natural attitude, Barrett posits that it is now possible to virtually recreate an exact replication (or synthesis) of sound fields in which our perception of a sound source (or sound object) localisation is as accurate as in naturally occurring sound fields[45].

This progressive acknowledgment of spatialisation technology by Barrett can be further defined in its movement away from traditional *per speaker channel* based approaches to spatialisation (such as stereophony, 5.1 surround and multi speaker diffusion), toward the encoding of digital audio files by defining spatial parameters stored as metadata. This approach is known as *object-oriented* spatialisation and is designed to allow the reproducibility of auditory spatial cues over an arbitrary array of speakers[46]. This technique is highly dependent on the encoding and decoding algorithms used and is in the early stages of development and standardisation. An example of the commercialisation of this technology can be seen with the new Dolby Atmos standard for film sound spatialisation[47]. The major strength of this approach from the perspective of electroacoustic musicians is that spatial ideas can be articulated more effectively to the listener over a vastly larger listening area independent from the sweet spot and which can be decoded accurately to an arbitrary speaker configuration independent of that used while composing. Interestingly this *object oriented* approach can be considered an *abstraction* of the "(spatialised) sound object" to exist

[45] Natasha Barrett, "Creating Sonic Spaces: An Interview with Natasha Barrett," in *Computer Music Journal*, no. 31:2 (2007), 12.
[46] Guillume Potard, "3D-audio object oriented coding," (Ph.D. diss, University of Wollongong, 2006), http://ro.uow.edu.au/theses/539
[47] Brett Crockett, "Dolby Atmos: Why object-based audio matters," *Lab Notes*, September 29, 2014, accessed May 28, 2015, http://blog.dolby.com/2014/09/dolby-atmos-object-based-audio-matters/

independent of technological limitations. In a way this may be considered analogous to, or at least influenced by the phenomenological attitude of Schaeffer.

Rising interest in this type of research has led to a project called CARUSSO (Creating, Assessing and Rendering in Real Time of High-Quality Audio-Visual Environment in MPEG-4 Context) whose aims have explicitly been to record, encode, transmit and reproduce sound fields. There currently exist file types that can facilitate the encoding of spatial information such as the MPEG 4 and XML3DAudio formats[48]. In essence these formats are trying to separate compositional techniques from reproduction technology so that complex electroacoustic hardware calibration is not so crucial to the success of the *transference* of a spatial composition.

The influence of technological development in electroacoustic spatialisation is particularly evident in the work of Marije Baalman. Primarily her research engages with the continuous discourse needed between spatial compositional considerations and the spatialisation technologies needed for implementation. Baalman states that in general, electroacoustic composers are very much used to working with stereophonic technology and as a result this forms the basis for their compositional ideas concerning space within that technology[49].

Baalman herself represents the situation of many contemporary electroacoustic musicians in that the basis of her interest in music is highly influenced by previous education in the physical sciences. Whether this awareness is achieved through an institutionally prescribed approach or by ones own means, making music using electroacoustic technology requires at least some understanding of the disciplines grounded in the natural attitude. Baalman's praxis

[48] R. Ranjan and W-S Gan. 2013. "Wave Field Synthesis: The Future of Spatial Audio," Accessed 2 September 2014, 22 http://ieeexplore.ieee.org.proxy.library.adelaide.edu.au/stamp/stamp.jsp?tp=&arnumber=6482237&tag=1
[49] Marije. A. Baalman, "Spatial Composition techniques and Sound Spatialisation Technologies," *Organised Sound,* 15, (2010), 209-218 doi: 10.1017/S1355771810000245

based research is pragmatic in objective and looks to highlight new compositional methodology based on the employment of the state of the art and the concurrent influence this technology imposes within this continuum[50]. This background provides another perspective with which to achieve a theorisation of spatio-musical concepts.

Research concerning these new speaker systems that enable the fine parametric control of the sound field is contributing significantly to contemporary discourse in electroacoustic music presentation. The types of control established through algorithmic development and the exponential rise of computing power (particularly in the consumer market) is making the utilization of these technologies feasible in a musical context[51].

Smalley has emphatically considered the fragility of superimposing a *composed space* within the *listening space*, and as such outlines a perceptual framework for the *transference of spatial articulation*. This issue has also been addressed from grounding in the natural attitude. The active electroacoustic control of room acoustics is a commercially established field in acoustics consultation and provides a workable solution to modify the acoustic response of sounding spaces in situations where architectural modifications are impractical[52]. These systems are a by product of research undertaken in sound field synthesis and have resided not only within physics departments of universities but in select practice rooms and concert halls for the better part of the last fifteen years[53]. Although not a fully parametric solution to the control of room acoustics, these technologies can offer the possibility of acute control of the physical listening space to the musician.

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[50] Marije. A Baalman, *On Wave Field Synthesis and Electro-acoustic Music*, Saarbruken, Germany: VDM verlag Dr Muller Aktiengesellschaft & Co., 2008, 11
[51] Abbott, 12
[52] John Matheson, "Theatre sound system enhances music experience," *Lares: What people said* (1999), accessed June 2, 2015, http://www.lares-lexicon.com/pdfs/AFCArticle.pdf
[53] Wenger VAE, wengercorp.com, accessed June 2, 2015, https://www.wengercorp.com/acoustics/vae-rehearsal.php

In an attempt to identify a definitive vocabulary for electroacoustic spatiality Kendall and Ardila list three reasons why the potential of spatialisation is rarely realized in a musical sense: 1) misconceptions about the technical aspects of spatialisation systems, 2) misconceptions about the nature of spatial perception, *especially in context of such systems*, and 3) a lack of creative engagement, possibly due to the first two issues[54].

[54] Gary S. Kendall and Mauricio Ardila, "The Artistic Play of Spatial Organisation: Spatial Attributes, Scene Analysis and Auditory Spatial Schemata," in *Computer Music Modeling and Retrieval: Sense of Sounds,* ed. Richard Kronland-Martinet, Sølvi Ystad and Kristoffer Jensen (Berlin: Springer-Verlag, Heidelberg, 2008), http://dx.doi.org/10.1007/978-3-540-85035-9_8, 126.

Conclusions

Both Spatial Articulation and subsequent theories for sound spatialisation see the means of achieving spatio-musical expression as a core consideration in presenting electroacoustic music to an audience. The primary distinction between both types of theory has been summarized through a critical analysis of theoretical frameworks relating to and including Smalley's Spatial Articulation. By juxtaposition of theory both grounded in the phenomenological attitude and the natural attitude, this study has highlighted a distinct dichotomy in perspective used by prominent theorists when addressing the use of sound spatialisation as a rudiment of electroacoustic music.

In adopting the phenomenological reduction of sound central to Schaefferian theory, Smalley provides a perceptual centric approach to electroacoustic music theory. This passive rejection of the natural standpoint or 'bracketing' in philosophical terms, is necessary in developing a coherently descriptive tool set with a view to addressing the notoriously complex conceptions of space inherent in electroacoustic composition and listening. In a summary of his framework Smalley concludes that space must be approached experientially and based on his assertions the composer cannot systematically control spatial experience in electroacoustic music. Of interest are his acknowledgments of holosonic control of the sound field (or current lack thereof) and as such highlights his bracketing of the natural attitude.

In contrast with Smalley's phenomenology, an understanding of technological development and how it can afford the accurate projection of spatialised sound has influenced other theorists. This is a complex field of enquiry which necessitates a competence in various disciplines including; applied physics (specifically acoustics), psychoacoustics, computer science and electronics theory. Here the aforementioned realist perspective (the natural attitude) for theoretical development can be seen to encapsulate these necessary disciplines

whose basis' are of the physical sciences. The emergence of words like *parameter*[55] and *parametric* in electroacoustic music discourse shows the tendency of more recent theorists to conceptualise spatialisation from the natural attitude standpoint. To determine whether this perspective is intentional or is a result of a passive socio-technical understanding relative to those who posited theories decades before them will require further research. However the increasing prevalence of this natural attitude shows an alternate grounding for electroacoustic music theory than that of the phenomenological attitude. As such the natural attitude is greatly influencing the current discourse surrounding space as a rudimentary aspect of electroacoustic music. Developments including; the object oriented approach to coding spatial variables within digital audio formats that can be articulated on arbitrarily defined speaker arrays, and the vast enlargement of the listening area in which spatially defined auditory cues remain stable, can be seen as a major rectification of the problem of *transference of spatial articulation* into the listening environment.

For Smalley an ad hoc experiential process is necessary to address electroacoustic spatialisation within the listening environment. This provides a mass of unquantifiable solutions for which he posits a descriptive tool set based on aural perception and the phenomenological attitude. For others the physical properties, and therefore the quantifiable parameters of the sound field provide a foundation for theoretical reasoning. As a result of technological developments in sound system engineering and consumer computing, the tools for enabling this acute variable control of space are within reach of the musician.
While each diametrically opposed grounding in theory provides substantial benefit to discourse around this complex aspect of electroacoustic music, further research into how each may complement one another seems of great importance. While Doornbusch and McIlwain have explicitly assumed the total parametric control of spatialised sound in any environment

[55] Paul Lansky and George Perle. "Parameter." *Grove Music Online. Oxford Music Online*. Oxford University Press. Web. 22 Jun. 2015. http://www.oxfordmusiconline.com/subscriber/article/grove/music/20880.

to focus upon phenomenological issues of space, Barrett and Baalman have both briefly referenced considerations from theoretical groundings opposite to theirs without an explicit acknowledgment of this difference in grounding. Kendall and Ardila define misconceptions about the nature of spatial perception as particularly evident in the context of considering new electroacoustic spatialisation systems. This idea could provide a platform for further research into the dichotomy of these two theoretical groundings in the phenomenological attitude and the natural attitude, and further what possible benefits can be established by a reintegration of studies in each vein.

> As the architect bases his structures on a perfect knowledge of the materials he uses -- their resistance, their reaction, their tensile strength - the composer today should, in building his sonorous constructions, have a thorough knowledge of the laws governing the vibratory system in the atmospheric domain, and off the possibilities that science has already abundantly placed, and continues to place, at the service of his imagination. The last word is: imagination. - Edgard Varese[56]

[56] Edgard Varèse, "Organized Sound for the Sound Film," in *The Commonweal* (1940): 204-05. Print.

Bibliography

Abbott Nathan. "Facilitating Spatial Music: assessing feasibility in the amateur implementation of advanced spatialisation technology." Undergraduate essay, University of Adelaide, 2014.

Baalman Marije. A. *On Wave Field Synthesis and Electro-acoustic Music*. Saarbruken, Germany: VDM verlag Dr Muller Aktiengesellschaft & Co., 2008.

Baalman Marije. A. "Spatial Composition techniques and Sound Spatialisation Technologies." *Organised Sound* (2010). accessed September 21, 2014. doi:10.1017/S1355771810000245

Barrett Natasha. "Spatio-musical composition strategies," in *Organised Sound* (2002). Accessed April 3, 2015. http://journals.cambridge.org.proxy.library.adelaide.edu.au/article_S1355771802003114

Barrett Natasha. "Creating Sonic Spaces: An Interview with Natasha Barrett," in *Computer Music Journal*, no. 31:2 (2007), 12.

Burt Warren. "Ways of Listening." accessed May 28, 2015. http://www.warrenburt.com/ways_of_listening/.

Chion Michel. *Guide des objets Sonores: Pierre Schaeffer et la recherché musicale*. Translated by John Dack and Christine North. Paris: Édition Bucket/Chastel, 1983.

Chion Michel. *Audio-Vision: Sound on Screen*. Columbia University Press, 1994.

Crockett Brett. "Dolby Atmos: Why object-based audio matters." *Lab Notes*, September 29, 2014. Accessed May 28, 2015. http://blog.dolby.com/2014/09/dolby-atmos-object-based-audio-matters/.

Doornbusch, P & McIlwain, P. "Integrating Spatial Parameters in Composition." Accessed September 5, 2014. https://www.academia.edu/168451/Integrating_Spatial_Parameters_in_Composition_Practice.

Emmerson Simon. "Introduction," in *The Language of Electroacoustic Music*. Edited by Simon Emmerson. Basingstoke: Macmillan Press, 1986.

Farnell Andy. *Designing Sound*. Cambridge, Massachusetts: The MIT Press, 2010.

Matheson John. "Theatre sound system enhances music experience." *Lares: What people said* 1999. Accessed June 2, 2015. http://www.lares-lexicon.com/pdfs/AFCArticle.pdf

Trochimczyk Maja. "Space and Spatialization in Contemporary Music: History and Analysis, Ideas and Implementations." Ph. D. Dissertation. Montreal, Quebec, Canada: McGill University, Faculty of Music, 1994. PDF Reprint. Los Angeles, California: Moonrise Press, 2011.

Kane Brian. “L’Objet Sonore Maintenant: Pierre Schaeffer, sound objects and the phenomenological reduction.” in *Organised Sound,* 2007. Accessed May17, 2015. http://journals.cambridge.org/abstract_S135577180700163X.

Kendall G and Ardila M “The Artistic Play of Spatial Organisation: Spatial Attributes, Scene Analysis and Auditory Spatial Schemata.” In *Computer Music Modeling and Retrieval: Sense of Sounds*. Edited by Richard Kronland-Martinet, Sølvi Ystad and Kristoffer Jensen. 126. Berlin: Springer-Verlag, Heidelberg, 2008. http://dx.doi.org/10.1007/978-3-540-85035-9_8.

Kim Suk-Jun. “A Critique on Pierre Schaeffer’s Phenomenological Approaches: Based on the Acousmatic and Reduced Listening.” Paper presented at the Pierre Schaeffer Conference: mediART in Rijeka, Croatia, on Oct. 7, 2010. http://reddoorsound.com/files/PS_Phenomenology_KIM.pdf.

Nattiez Jean-Jacques. *Music and Discourse: Towards a Semiology of Music*. Princeton: Princeton University Press, 1990.

Perle G and Lansky P. "Parameter." *Grove Music Online*. *Oxford Music Online*. Oxford University Press. Web. 22 Jun. 2015. http://www.oxfordmusiconline.com/subscriber/article/grove/music/20880.

Potard Guillume. “3D-audio object oriented coding.” Ph.D., dissertation, University of Wollongong, 2006. http://ro.uow.edu.au/theses/539.

Reed Tim. “I. Planes of discourse in fixed media electroacoustic music: A comparative study and application of analytical approaches, and II. Three movements for string orchestra.” Ph.D., dissertation, University of Florida, 2008. Accessed May 10, 2015. http://proxy.library.adelaide.edu.au/login?url=http://search.proquest.com.proxy.library.adelaide.edu.au/docview/250785332?accountid=8203

Smalley Denis. “Spectro-morphology and Structuring Processes.” In *The Language of Electroacoustic Music*. Edited Simon Emmerson. Basingstoke: Macmillan Press, 1986.

Smalley Denis. “Spatial Experience in Electroacoustic Music.” In *L’Espace du Son II*. Edited by Francis Dhomont. Ohain: Musiques et Recherché, 1991. Accessed April 8, 2015. http://www.musiques-recherches.be/images/stories/eds2.pdf.

Smalley Denis. “Spectromorphology: explaining sound-shapes.” In *Organised Sound,* 1997. Accessed May 20, 2015. http://journals.cambridge.org/abstract_S1355771897009059.

Smalley Denis. “The Listening Imagination: listening in the electroacoustic era.” In *Contemporary Music Review* (2009). Accessed April 8, 2015. http://dx.doi.org/10.1080/07494469600640071

Tuuri K and Eerola T. “Formulating a Revised Taxonomy for Modes of Listening.” In *Journal of New Music Research* (2012). Accessed May 28, 2015. http://www.tandfonline.com.proxy.library.adelaide.edu.au/doi/citedby/10.1080/09298215.2011.614951#.VYdtncYsojI.

Varèse Edgard. "Organized Sound for the Sound Film." In *The Commonweal* (1940): 204-05.

Weale Rob. “Electroacoustic Music: English Definition.” Electroacoustic resource site. Accessed May 2, 2015. http://ears.pierrecouprie.fr/spip.php?article109

“Wenger VAE.” wengercorp.com. Accessed June 2, 2015. https://www.wengercorp.com/acoustics/vae-rehearsal.php.

Worrall, David. “Space in sound: sound of space.” In *Organised Sound*, 2000. Accessed September 2, 2014. http://dx.doi.org.proxy.library.adelaide.edu.au.